

# ALIAH UNIVERSITY

## End Semester Examination (Spring Semester)2026 (B. Tech 3<sup>rd</sup> Year 6<sup>th</sup> Semester)

Subject Name: Software Engineering

Full Marks: 80

Time: 3 Hours

Subject Code: CSEUGPC17

### Group A Answer all 5 questions

1. Answer the following with at most 2 sentences. 5 X 2= 10
- a) What is meant by software process model? (CO1 - Remember)
  - b) Define cohesion in software design. (CO2 - Understand)
  - c) What is cost estimation in software engineering? (CO4 - Understand)
  - d) What is software testing? (CO3 - Understand)
  - e) Define risk in software projects. (CO4 - Understand)

### Group- B Answer any 6 questions

6 X 5 = 30

- 2. Explain equivalence class partitioning with an example. (CO3 - Understand)
- 3. Describe Expert Judgement and Delphi estimation techniques. (CO4 - Understand)
- 4. Explain Function Point metric and its advantages over LOC. (CO1 - Understand)
- 5. Discuss why Spiral model is considered a meta-model. (CO1 - Understand)
- 6. A project estimated using Putnam model takes 10 months and costs ₹80,000. Estimate new cost if time is reduced to 5 months. (CO4 - Apply)
- 7. Explain boundary value analysis with suitable examples. (CO3 - Understand)
- 8. Describe critical path and critical tasks with diagram. (CO4 - Understand)

### Group- C Answer any 4 questions

4 X 10 =40

- 9. a) Using black box testing, design test cases for a program that checks whether a number is even or odd. (CO3 - Apply)
- b) Explain Alpha, Beta and Acceptance testing. (CO3 - Understand) 5+5
- 10. a) Explain Basic and Intermediate COCOMO models. (CO4 - Understand)

- b) Describe Complete COCOMO model. (CO4 - Understand) 5+5
11. a) Describe different types of coupling. (CO2 - Understand)
- b) A software size is estimated as 40,000 LOC. Average salary = ₹60,000/month. Estimate effort, development time and cost. (CO4 - Apply) 5+5
12. a) Explain all phases of Waterfall model with diagram. (CO1 - Understand)
- b) Discuss advantages and limitations of Waterfall model. (CO1 - Analyze) 5+5
13. a) Write pseudocode for factorial of a number. (CO3 - Create)
- b) Draw CFG and calculate cyclomatic complexity. (CO3 - Create) 5+5

# Aliah University

End-Semester Examination - 2026

(BTech, CSE 3<sup>rd</sup> Year, 6<sup>th</sup> Semester)

Paper Name: Compiler Design

Full Marks: 80

Paper Code: CSEUGPC18

Time: 3 Hrs

## Group - A

(10X1=10)

### 1. Fill in the blanks -

[Remember, Understand, CO<sub>1</sub>, CO<sub>2</sub>, CO<sub>3</sub>, CO<sub>4</sub>]

- 'R' in LR parser stands for \_\_\_\_\_.
- For tokenization, \_\_\_\_\_ is better than Buffer-pair concept.
- Context Free Grammars are used in \_\_\_\_\_ phase of compilation.
- Instead of 'a\*2', writing 'a+a' is optimized code, this approach is named as \_\_\_\_\_.
- Regular expression of words with 'a' to 'm' inputs is \_\_\_\_\_.
- In contrast with AST, DAG never repeats \_\_\_\_\_.
- Inputs of lexical analyzer are called \_\_\_\_\_.
- Information of any token is stored in \_\_\_\_\_.
- A CFG is ambiguous if there exists more than one \_\_\_\_\_.
- 'Array out of bound' exception could be detected in \_\_\_\_\_ phase of compilation.

## Group - B

Answer any five of the following questions -

(5X6=30)

- Write a lex code to count number of vowels and non-vowel from inputs. [Remember, CO<sub>1</sub>]
- Write an SDT to convert any infix expression into postfix expression. [Apply, CO<sub>4</sub>, CO<sub>5</sub>]
- Write short notes on AST and DAG, and form AST and DAG for the following statement –  
$$P = a + (b * c) + (d - e) * (b * c)$$
 [Apply, CO<sub>4</sub>]
- Mention the minimum requirements for a grammar to be treated as LL(1) parser. If any grammar doesn't satisfy the conditions, then what are various steps you should take to make it 'ready' to be treated as LL(1) parser? [Analyze, CO<sub>4</sub>]
- Define and discuss about S-Attributed and L-Attributed Syntax Directed Translations with the help of few examples. [Understand, CO<sub>4</sub>, CO<sub>5</sub>]
- Write a Context Free Grammar for the Language below –  
$$L: \{a^n b^m c^m d^m \mid n, m > 0, \Sigma = \{a, b, c, d\}\}$$
 [Apply, CO<sub>2</sub>, CO<sub>3</sub>]
- Write regular expressions and draw Deterministic Finite Automata for the following tokens  
- identifiers, integer constants, floating constants. [Apply, CO<sub>1</sub>, CO<sub>2</sub>]

**Group - C**

**Answer any four of the following questions -**

**(10X4=40)**

9. a) Mention different roles of Lexical Analyzer. Explain Buffer pair concept, used for Tokenization. [Analyze, CO2] [3+3]  
b) Construct a Deterministic Finite Automata to accept the following language- [Evaluate, CO2] [4]  
**L: All the strings of at-least 3 length and starting with 'a', where  $\Sigma = \{a,b\}$**

10. What are various ways to optimize a loop? Briefly explain other machine independent code optimization techniques. Mention some 'peephole optimization' techniques too. [4+3+3]  
[Understand, CO5]

11. Calculate First() and Follow() for all the non-terminals of the grammar given below. Form a predictive parsing table and conclude whether the following grammar could be considered as LL(1) parser or not. [Evaluate, CO3, CO4] [3+3+3+1]

$S \rightarrow iEtSA \mid a$

$A \rightarrow eA \mid \epsilon$

$E \rightarrow d$

Here  $N = \{S, A, E\}$  and S is the start symbol

12. Calculate LR (0) item sets for the grammar given below. Then, construct SLR(1) parsing table. [Evaluate, CO4, CO5] [6+4]

$S \rightarrow AA$

$A \rightarrow aA$

$A \rightarrow b$

Here  $N = \{S, A\}$  and S is the start symbol

13. a) Write 3-address code for the c-program segment given below. [Apply, CO4] [4]

```
for (i = 0; i < 10; i++) {  
    if (a < b) {  
        x = y + z;  
    } else {  
        x = y - z;  
    }  
}
```

- b) Represent the above 3-address code in Quadruples and Triples format. [Apply, CO4] [3+3]
-

**Aliah University**  
**End Semester Examination (Spring Semester) 2026**  
**(For 3<sup>rd</sup> Year 6<sup>th</sup> Semester B.Tech(CSE))**

**Paper Name: Embedded Systems**  
**Paper Code: CSEUGPC21**

**Full Marks:80**  
**Time:3 hours**

**Group A (Answer all questions)**

**10X1=10**

1. What is Embedded System? [CO1, Remember]
2. Differentiate event-triggered and time-triggered system. [CO2, Understand]
3. Give examples of embedded system used in health care. [CO1, Remember]
4. Give examples of some fourth-generation embedded system. [CO6, Remember]
5. Differentiate sensor and actuator. [CO2, Understand]

**Group B (Answer any 5 questions)**

**6X5=30**

1. Differentiate
  - a) I2C and SPI
  - b) Microprocessors and Microcontrollers (3+3) [CO4, Understand]
2. What are the characteristics of Embedded System? [CO2, Remember]
3. Explain the different components that constitute the core of an embedded system. [CO3, Understand]
4. Explain how Embedded System function in the Data Communication domain. [CO6, Understand]
5. Differentiate Hard and Soft Real Time System. [CO3, Understand]
6. Differentiate Serial and Parallel Communication Interface. [CO3, Understand]

**Group C (Answer any 4 questions)**

**(4X10=40)**

1. a) Explain different classifications of embedded system with examples. [CO1, Understand]
  - b) Explain the different operational quality attributes in detail. (5+5) [CO2, Understand]
2. a) Compare the operations of Zigbee and Wi-Fi networks. [CO3, Understand]
  - b) Discuss about the memory unit used in Embedded Systems. (5+5) [CO3, Understand]
3. a) Explain Rate Monotonic Scheduling Algorithm. [CO2, Understand]
  - b) Schedule the following processes by Rate Monotonic Scheduling.

(5+5)

| Processes | Execution Time (C) | Time period (T) |
|-----------|--------------------|-----------------|
| P1        | 3                  | 20              |
| P2        | 2                  | 5               |
| P3        | 2                  | 10              |

[CO5, Apply]

4. Write short notes on

(2.5X4)

[CO3, Understand]

- a) Digital Signal Processor
- b) Harvard Architecture
- c) FPGA
- d) UART

5. a) Define with examples the two types of Communication Interfaces.

[CO1, Remember]

b) Explain the sequence of operation for communicating with I2C slave device with the help of a diagram.

[CO5, Understand]

(3+7)

**Paper Name: Computer Networks**  
**Paper Code: CSEUGPC19**

**Group A**

I. Which error detection method can correct single-bit errors? [CO1 Remembering]  
a) CRC                  b) Hamming Code                  c) Checksum      d) Parity

III. Which protocol is used to assign IP addresses dynamically? [CO1 Remembering]  
a) DNS                      b) DHCP                      c) FTP                      d) SNMP

V. Which multiple access technique uses time slots? [CO2 Understanding]  
a) FDMA                      b) TDMA                      c) CDMA                      d) ALOHA

**VII.** TCP connection establishment uses: [CO1 Remembering]  
a) 2-way handshake ~~b) 3-way handshake~~ c) 4-way handshake d) Stop-and-wait

**IX.** What is the main purpose of TTL field? [CO2 Understanding]

a) Error detection                      ~~b) Prevent infinite looping~~  
c) Flow control                          d) Congestion control

X. CDMA is based on: [CO2 Understanding]  
a) Time division b) Frequency division c) Code division d) Space division

**Group B**  
**Answer any 4 question**

**4X5=20**

2. Compare Sliding Window Protocol and Stop & Wait ARQ. [CO4 Analyzing]
3. Differentiate between Circuit Switching and Packet Switching. [CO5 Evaluating]
4. Explain different persistence methods used in CSMA. [CO2 Understanding]
5. Compare ARP and RARP. [CO2 Understanding]
6. Describe different types of HDLC frames with functions. [CO2 Understanding]
7. Write a short note on CRC and its working principle. [CO2 Understanding]

**Group C**  
**Answer any 5 question**

**5X10=50**

8.
  - a. Draw and explain IPv4 header format.
  - b. Briefly describe the working principle of ARP. [CO3] Applying + CO2 Understanding (6+4=10)
9. Represent the bit stream 101101001 using:
  - a. NRZ
  - b. NRZ-L
  - c. NRZ-I
  - d. Manchester Encoding
  - e. Differential Manchester Encoding [CO3 Applying] (2x5=10)
10. You are given IP address 10.0.0.0. You need to create 1025 subnets.
  - a. Determine the new subnet mask  $2^{35} \cdot 2^{55} \cdot 2^{24} \cdot 0$
  - b. Number of hosts per subnet  $2^{13} \cdot 2$
  - c. Broadcast address of 10th subnetwork  $10 \cdot 1 \cdot 63 \cdot 255$
  - d. Network address of last subnetwork  $10 \cdot 255 \cdot 224 \cdot 0$
  - e. IP address of 25th host in last subnetwork  $10 \cdot 255 \cdot 224 \cdot 25$  [CO3] Applying (5x2=10)
11.
  - a. Explain Token Bucket algorithm with diagram.
  - b. Explain the role of Poison Reverse in routing. [CO3 Understanding / Applying] (6+4=10)
12.
  - a. Explain CSMA/CA with suitable diagram.
  - b. Compare CSMA/CD and CSMA/CA. [CO3 Applying] (6+4=10)
13. Write short notes on any two:
  - a. NAT and PAT
  - b. DNS
  - c. Ping
  - d. Bellman Ford Algorithm [CO4 Analyzing] (2x5=10)



# Aliah University

Even-Semester (Spring) Examination - 2026

(For 6<sup>th</sup> Semester BTech. CSE)

Paper Name: Computer Graphics

Full Marks: 80

Paper Code: CSEUGPC20

Time: 3 Hrs

(Answer the Questions as per the instruction in each Group and in Sequential Order)

## Group - A

1. Answer all the questions:

(10X1=10)

| SL No. | Questions                                                                                                                                                    | Marks |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| a)     | Which display device works on the principle of electron beam scanning?<br>a) LCD b) LED <del>c) CRT</del> d) Plasma                                          | 1     |
| b)     | The smallest unit of a digital image is called:<br>a) Byte <del>b) Pixel</del> c) Vertex d) Frame                                                            | 1     |
| c)     | The DDA algorithm is mainly used for:<br>a) Polygon filling b) Clipping <del>c) Line generation</del> d) Shading                                             | 1     |
| d)     | Which transformation changes the size of an object?<br>a) Rotation b) Translation <del>c) Scaling</del> d) Reflection                                        | 1     |
| e)     | A point (x,y) is translated by (h,k). The correct formula is:<br>a) $(x',y')=(x-h,y-k)$ b) $(x',y')=(x+h,y+k)$ c) $(x',y')=(xh,yk)$ d) $(x',y')=(x/k,y/h)$   | 1     |
| f)     | Reflection about the X-axis changes:<br>a) x-coordinate only b) y-coordinate only c) both x and y coordinates<br>d) neither x nor y coordinates              | 1     |
| g)     | Windowing in computer graphics is used to:<br>a) Display multiple monitors <del>b) Select a part of a scene</del><br>c) Increase resolution d) Reduce memory | 1     |
| h)     | Which technique is used to fill a polygon area with color?<br>a) Clipping <del>b) Flood Fill</del> c) Scaling d) Reflection                                  | 1     |
| i)     | The Cohen-Sutherland algorithm uses:<br><del>a) Region codes</del> b) Control points c) Pixel masking d) Texture mapping                                     | 1     |
| j)     | Which transformation keeps the shape unchanged but changes its position?<br><del>a) Translation</del> b) Scaling c) Shearing d) Reflection                   | 1     |

## Group - B

(Answer any 6 questions)

(6X5=30)

| Sl. No. | Questions                                                                                                        | Marks |
|---------|------------------------------------------------------------------------------------------------------------------|-------|
| 1. a)   | Define computer graphics.                                                                                        | 1+4   |
| b)      | Draw the different component of computer graphics system and explain each part.                                  |       |
| 2.      | Explain the working procedure of CRT monitor with proper diagram.                                                | 5     |
| 3.      | Write short note on any two input device.                                                                        | 5     |
| 4.      | Consider a line with endpoints (20,10) and (30,18). Use Bresenham line drawing algorithm to rasterize this line. | 5     |
| 5.      | Define translation and rotation using matrix form.                                                               | 2x2.5 |
| 6.      | Explain different types of line caps and joins with diagram.                                                     | 2x2.5 |

|    |                                                                                                                               |   |
|----|-------------------------------------------------------------------------------------------------------------------------------|---|
|    |                                                                                                                               |   |
| 7. | Translate the triangle with vertices A(1,2),B(4,2),C(2,5) by the vector (-2,3). Find the new vertices. Also draw the diagram. | 5 |
| 8. | What is a composite transformation? Explain successive translation is additive.                                               | 5 |

Group – C  
(Answer any 4 questions)

(10X4=40)

| Sl No. | Questions                                                                                                                                                                 | Marks |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1.     | Write down and derive the DDA line drawing algorithm. What is the advantages and disadvantages of it.                                                                     | 6+4   |
| 2.     | Calculate the pixels for the first octant of circle for radius =10 pixels with its centre at origin? Also show the pixels for 2 <sup>nd</sup> and 3 <sup>rd</sup> octant. | 7+3   |
| 3.     | Use the Cohen Sutherland algorithm to clip line P1(70,20) and P2(100,10) against a window A(50, 10) B(80,10) C(80,40) D(50,40) with diagram.                              | 10    |
| 4.     | Rotate a triangle A(0,0), B(2,2), C(4,2) about the origin and about P(-2,-2) by an angle of 45 degree.                                                                    | 10    |
| 5.     | Explain Sutherland-Hodgeman polygon Clipping algorithm with suitable example.                                                                                             | 10    |
| 6.     | What is RGB color model? Differentiate between RGB and Gray color model .                                                                                                 | 5+5   |

**End-Semester Examination (Spring Semester) - 2025-26****[B.Tech, CSE - 3rd Year, 6th Semester]**

Paper Name: Data Science & Big Data  
 Paper Code: CSEUGPE02

Full Marks: 80  
 Time: 3 Hrs

**GROUP A****[Answer any five questions]****[5 × 2 = 10]**

1.

- i) What is meant by data reduction in the data-preprocessing stage?
- ii) Define support count of an itemset in association-rule mining.
- iii) If a frequent itemset contains four items, how many non-empty association rules can be generated from it?
- iv) State one reason for choosing a medoid instead of a mean point as a cluster representative.
- v) Compute the z-score of  $x = 86$  when the mean is 68 and the standard deviation is 6.
- vi) Distinguish between feature selection and feature extraction.

**GROUP B****[Answer any four questions]****[4 × 5 = 20]**

- 2. Carry out two iterations of K-means clustering for the points (1,3), (2,4), (3,2), (8,7), (9,9) and (10,8). Use initial centroids  $C_1 = (1,3)$  and  $C_2 = (10,8)$ , and show the assignments and updated centroids after each iteration.
- 3. Explain the Apriori algorithm for association-rule mining. Include candidate generation, pruning and the use of the downward-closure property.
- 4. Classify nominal, ordinal, interval and ratio attributes with suitable examples taken from a student-performance dataset.
- 5. For the following confusion matrix, compute accuracy, precision, recall, false positive rate, and F-measure.

|                         | Actually Defective | Not Actually Defective |
|-------------------------|--------------------|------------------------|
| Predicted Defective     | 126                | 18                     |
| Predicted Not Defective | 9                  | 87                     |

- 6. A training set has three features (A, B, C) and two classes 'Green' and 'Orange': {(8.9, 84, 8.1, Green), (8.2, 79, 7.4, Green), (9.4, 88, 8.5, Green), (6.0, 50, 5.2, Orange), (5.6, 45, 4.7, Orange), (6.8, 56, 5.8, Orange), (9.1, 92, 8.9, Green), (6.1, 48, 5.0, Orange)}. Using KNN with  $k = 5$ , predict the class of (6.5, 54, 5.5).

**GROUP C****[Answer any five questions]****[5 × 10 = 50]**

- 7. a) Compute the entropy of a dataset in which 65% of the records are labeled Yes and 35% are labeled No.
- b) From the following distance matrix, draw the dendrogram using agglomerative complete-linkage clustering.

**[3 + 7]**

|   | 1   | 2   | 3   | 4   | 5   |
|---|-----|-----|-----|-----|-----|
| 1 | 0.0 |     |     |     |     |
| 2 | 3.6 | 0.0 |     |     |     |
| 3 | 5.9 | 4.4 | 0.0 |     |     |
| 4 | 8.8 | 7.2 | 3.7 | 0.0 |     |
| 5 | 9.5 | 8.1 | 4.6 | 2.9 | 0.0 |

8. a) Consider the transaction database below. For minimum support = 3, construct the FP-growth tree.

$t_1: b, d$   $t_2: a, b, c$   $t_3: b, c, e$   $t_4: a, b, c, e$   $t_5: a, c, e$   $t_6: a, b$   $t_7: a, b, c, d, e$

b) Extract the frequent patterns from the FP-growth tree obtained in part (a).

[5 + 5]

9. a) Apply the Apriori algorithm to find all frequent itemsets from the following transactions for minimum support value 3.

$t_1: \underline{u}, \underline{v}, w$   $t_2: \underline{v}, \underline{x}, w$   $t_3: \underline{u}, \underline{y}$   $t_4: \underline{u}, \underline{v}, \underline{x}, \underline{y}, w$   $t_5: \underline{y}, z$   $t_6: \underline{u}, \underline{v}, \underline{y}$   $t_7: \underline{v}, w, x$

b) Define lift and explain how it helps to judge an association rule beyond support and confidence.

[7 + 3]

10. Consider the dataset  $\{(2, 6), (4, 4), (5, 8), (7, 7), (9, 10), (11, 9)\}$ .

a) Compute the covariance matrix for the dataset.

b) Determine the eigenvalues and corresponding unit-length eigenvectors of the covariance matrix.

[4 + 6]

11. a) Project the dataset given in Q10 onto a single principal component using Principal Component Analysis.

b) Consider the objects  $P_1(1, 1)$ ,  $P_2(2, 3)$ ,  $P_3(4, 2)$ ,  $P_4(8, 8)$ ,  $P_5(9, 7)$  and  $P_6(11, 9)$ . Build the Euclidean-distance weighted graph, compute its minimum spanning tree using Prim's or Kruskal's method, and apply MST-cut based divisive hierarchical clustering by deleting the two largest MST edges to obtain three clusters.

[4 + 6]

12. a) Explain the Partitioning Around Medoids (PAM) algorithm and mention why medoids are less affected by outliers than centroids.

b) For the objects  $A(2, 3)$ ,  $B(3, 5)$ ,  $C(4, 4)$ ,  $D(9, 8)$ ,  $E(10, 10)$  and  $F(11, 9)$ , take  $k = 2$  with initial medoids  $A$  and  $D$ . Using Manhattan distance, assign objects to the nearest medoid, perform one complete PAM swap check, and report the final medoids, total cost and clusters.

[4 + 6]